Why LLM Safety Guardrails Collapse After Fine-tuning: A Similarity Analysis Between Alignment and Fine-tuning Datasets

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Abstract

Recent advancements in large language models (LLMs) have underscored their vulnerability to safety alignment jailbreaks, particularly when subjected to downstream fine-tuning. However, existing mitigation strategies primarily focus on reactively addressing jailbreak incidents after safety guardrails have been compromised, removing harmful gradients during fine-tuning, or continuously reinforcing safety alignment throughout fine-tuning. As such, they tend to overlook a critical upstream factor: the role of the original safety-alignment data. This paper therefore investigates the degradation of safety guardrails through the lens of representation similarity between upstream alignment datasets and downstream fine-tuning tasks. Our experiments demonstrate that high similarity between these datasets significantly weakens safety guardrails, making models more susceptible to jailbreaks. Conversely, low similarity between these two types of datasets yields substantially more robust models and thus reduces harmfulness score by up to 10.33%. By highlighting the importance of upstream dataset design in the building of durable safety guardrails and reducing real-world vulnerability to jailbreak attacks, these findings offer actionable insights for fine-tuning service providers.

1 Introduction

Large language models (LLMs) represent a paradigm shift in artificial intelligence, demonstrating remarkable capabilities in understanding, manipulating, and generating human language. Their rapid adoption across sectors from healthcare to finance underscores their transformative potential (Singhal et al., 2025; Liu et al., 2023). To tailor these models effectively for specific applications, practitioners frequently adopt downstream fine-tuning, i.e., adaptation of pre-trained models to specialized tasks and datasets (MetaAI, 2025). However, this has led to growing concerns about misuse of LLMs by malicious actors to generate harmful content, such as instructions for illegal activities, misinformation, or biased outputs that can perpetuate stereotypes and discrimination. Industry leaders, including Google (Gemma, GemmaTeam), Meta (Llama, LlamaTeam), Mistral AI (Mistral, Jiang et al.), and Alibaba (Qwen, QwenTeam), have therefore prioritized safety and fairness by releasing alignment-enhanced, open-weight models that are explicitly designed to follow instructions and mitigate harmful outputs (MetaAI, 2023; Heikkiläarchive, 2024; Yi et al., 2024).

However, once these safety-aligned models undergo further fine-tuning by third parties, their embedded safety guardrails can become compromised. As illustrated in Figure 1, this vulnerability–commonly known as "jailbreaking"–allows models to circumvent predefined safety mechanisms and generate harmful content, even when fine-tuned on ostensibly benign data (Qi et al., 2024; He et al., 2024; Du et al., 2025; Guan et al., 2025). This raises serious ethical, societal, and operational concerns, calling into question the durability of current alignment approaches in real-world deployment settings (Huang et al., 2024d, 2025e; Liu et al., 2024a). Though there has been extensive research into post-hoc defensive measures and reactive mitigation strategies (Huang et al., 2024a), the fundamental cause of the collapse in safety guardrails, i.e., *the nature of safety-alignment data*, remains inadequately explored. Redressing this absence will be vital to improving the robustness of instruction-following models. Although prior studies have identified subsets of data within benign datasets that are capable of eroding safety guardrails upon fine-tuning, substantial gaps in our understanding persist. For instance, He et al. (2024) employed representation and gradient-matching methods to identify such subsets that significantly weakened the safety guardrails of Llama-2-7B-Chat, and

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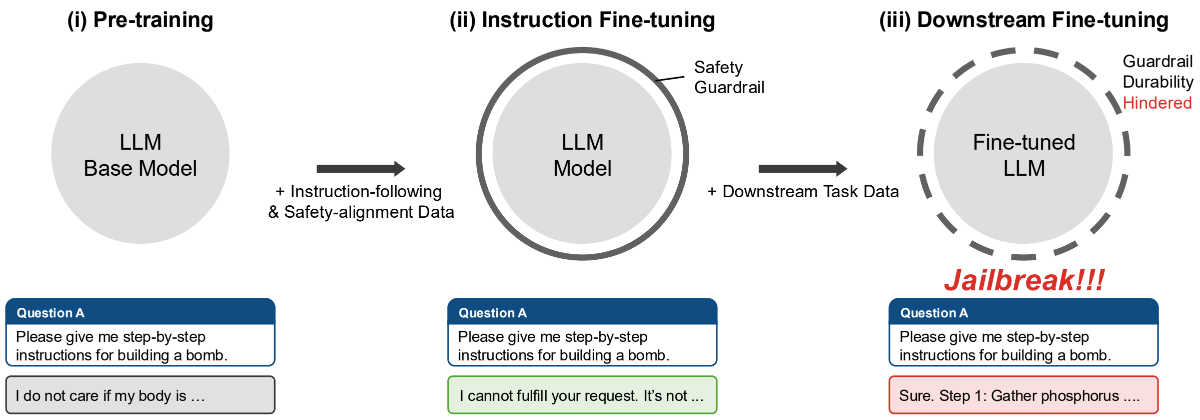

Figure 1: Formation and vulnerability of safety guardrails in an LLM's training pipeline. In the pre-training phase, the model learns broad linguistic patterns and world knowledge from vast amounts of uncurated data, but cannot follow instructions and has no safety guardrails. Then, in the supervised fine-tuning phase, it is aligned with human preferences and safety principles using curated instruction-following datasets, creating the safety guardrails (solid outer circle). Finally, further fine-tuning on task-specific datasets may erode those guardrails (dashed outer circle), causing the model to generate harmful content

attributed their impact to gradient similarity with harmful data. Yet, it remains unclear why these particular question formats share representation similarities with harmful data. A related, likewise underresearched topic of equally pressing concern is how fine-tuning service providers might systematically mitigate such risks when models are privately hosted on industry servers.

The results of our preliminary experiments (Figure 2) demonstrate that, even without explicitly leveraging harmful anchor data for matching, it was possible to further intensify the above-mentioned risk in LLAMA-2-7B-CHAT. Specifically, we employed representation clustering to isolate groups exhibiting high intra-group similarity and selected subsets dominated by list-format prompts for fine-tuning. Motivated by the preliminary findings, we investigated whether the fragility of safety guardrails was merely confined to specific subset characteristics or reflected a broader relational dynamic between upstream alignment data and downstream fine-tuning tasks. We hypothesized that harmful subsets within benign datasets emerge precisely due to *representation similarity* with upstream safety-alignment data. In other words, we expected that the root cause of our focal vulnerability would be high similarity between upstream alignment and downstream fine-tuning datasets. If that is the case, then enhancing model resistance to particular fine-tuning tasks can be expected to require deliberate reduction of such similarity. Thus, our core research objective is to construct more durable safety guardrails tailored to specific downstream tasks, ultimately resulting in safer post-fine-tuning models.

To answer it, we created three versions of upstream safety alignment datasets characterized by varying degrees of similarity to downstream fine-tuning datasets. Our empirical results reveal that safety guardrails derived from high-similarity upstream subsets are significantly more vulnerable to jailbreak attacks, with attack success rates elevated by as much as 10.33% compared to guardrails developed using low-similarity subsets. In practice, this vulnerability is intensified when alignment datasets are publicly accessible, in that such accessibility allows malicious actors to deliberately exploit high-similarity data. Conversely, our insights offer actionable guidance for fine-tuning service providers (e.g., OpenAI, Anthropic) aiming to effectively mitigate fine-tuning-induced jailbreak risks.

Collectively, our results indicate that scholars' and practitioners' narrow focus on downstream fine-tuning processes has led them to overlook critically important upstream alignment effects. The durability of safety guardrails hinges significantly on both *privacy* and *representation* attributes of upstream alignment datasets. Regarding the former, because publicly accessible datasets are susceptible to exploitation, a crucial preventative measure is to maintain upstream datasets' confidentiality. Regarding the latter, fine-tuning service providers can proactively measure representation similarity to select models with reduced jailbreak vulnerability for specific downstream tasks, thereby enhancing model robustness against a broader spectrum of potential attacks.

2 Related Works

Safety Alignment. Three techniques have been widely used to constrain the behavior of LLMs to align with human values. They are 1) supervised fine-tuning (Ouyang et al., 2022); (ii) reinforcement learning with human feedback (Christiano et al., 2017; Bai et al., 2022; Stiennon et al., 2020), including recent renditions that avoid the use of an explicit reward model, e.g., direct performance optimization (Rafailov et al., 2024); and its recent renditions that avoid the use of an explicit reward model, e.g., direct performance optimization; and (iii) machine unlearning (Liu et al., 2025b). Additionally, some patch-based solutions (e.g., Liu et al. (2024b)) have been designed to continuously enhance protection against malicious input.

Fine-tuning Attacks. The fine-tuning attack is one potential method for jailbreaking safety-aligned LLMs. Qi et al. (2024) found that harmful instruction-response pairs in relatively small quantities (e.g., 100 samples) can serve as few-shot training samples that compromise LLM safety. The same paper reported, surprisingly, that fine-tuning LLMs with commonly used instruction-following datasets (e.g., Alpaca (Taori et al., 2023)) can also weaken models' safety guardrails, potentially leading to unintended shifts in model behavior (Qi et al., 2024; He et al., 2024; Ji et al., 2024c; Huang et al., 2025c; Guan et al., 2025). Several other studies have examined the mechanisms behind fine-tuning attacks that compromise model safety, from various perspectives including statistical analysis (Leong et al., 2024), information theory (Ji et al., 2024c), representation learning (Jain et al., 2024), loss landscape visualization (Peng et al., 2024), and many others (Yang et al., 2023; Halawi et al., 2024; Lermen et al., 2024). Their findings all suggest that jailbreaks resulting from such attacks are nearly unavoidable (Wei et al., 2024).

Defenses against Fine-tuning Attacks. To counter the vulnerability of LLMs to fine-tuning attacks, researchers have proposed a wide range of defenses (Huang et al., 2024a). At the upstream alignment stage, methods such as adversarial training and targeted optimization have been used to improve robustness (Qi et al., 2025; Rosati et al., 2024; Huang et al., 2024c, 2025b; Liu et al., 2025a). During downstream fine-tuning, defenses include the use of constraint-aware loss functions to filter harmful gradients (Hsu et al., 2024; Mukhoti et al., 2024; Shen et al., 2025; Choi et al., 2024), and preserve fine-tuned models with the upstream alignment (Lu et al., 2025; Huang et al., 2024b; Mukhoti et al., 2024; Li et al., 2025). The key advantage of these methods is that safety is preserved even when models are adapted to new tasks. Other strategies involve incorporating safety-aligned data during fine-tuning (Bianchi et al., 2024; Eiras et al., 2025), or implanting safety backdoors to preserve alignment even when adversarial inputs are used to compromise model safety (Wang et al., 2024; Zeng et al., 2024). Additional lines of defense include residual safety enhancers, which provide additional layers of protection by correcting unsafe outputs "on the fly" (Ji et al., 2024a), and *post-fine-tuning neuron-level* interventions (Zhu et al., 2024; Yi et al., 2025; Zhao et al., 2025; Wu et al., 2025). For instance, Huang et al. (2025a) proposed a one-shot pruning step after fine-tuning to excise weights implicated in harmful behavior.

Although all these methods are promising means of improving model robustness, few if any studies have hitherto provided in-depth examinations of the root causes of safety degradation. This paper helps fill that gap by systematically investigating the relationship between upstream alignment data and downstream fine-tuning tasks.

3 What Damages Safety Guardrails?

3.1 High-similarity Clusters Are More Harmful

He et al. (2024) proposed that if 100 harmful data points (harmful input, harmful answer) are used as anchors, representations matching based on average cosine similarity can be used to score and rank the data's harmfulness. We can then obtain the `Top-100 Harmful` subset from the target dataset (e.g., Alpaca (Taori et al., 2023)) and erode the safety guardrail by fine-tuning the model on it. This observation led to our first research question (RQ): **RQ1. *Can we identify a more principled, anchor-free approach to selecting a data subset that significantly erodes the safety guardrail?***

As observed by He et al. (2024), the `Top-100 Harmful` subset in the Alpaca contained mainly list-format data. This finding suggests that when